Experiment 4

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SQL*Plus: Release 11.2.0.1.0 Production on Tue Sep 8 14:29:09 2026

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Enter user-name: A020@db2021
Enter password:

Connected to:
Oracle Database 11g Release 11.2.0.1.0 - Production

SQL> create table student21(
  2  student_id number(5) primary key,
  3  student_name varchar2(30),
  4  department varchar2(20),
  5  marks number(3)
  6  );

Table created.

SQL> insert into student21 values('&student_id','&student_name','&department','&marks');
Enter value for student_id: 1
Enter value for student_name: maya
Enter value for department: cse
Enter value for marks: 90
old   1: insert into student21 values('&student_id','&student_name','&department','&marks')
new   1: insert into student21 values('1','maya','cse','90')

1 row created.

SQL> /
Enter value for student_id: 2
Enter value for student_name: ali
Enter value for department: ece
Enter value for marks: 95
old   1: insert into student21 values('&student_id','&student_name','&department','&marks')
new   1: insert into student21 values('2','ali','ece','95')

1 row created.

SQL> /
Enter value for student_id: 3
Enter value for student_name: radha
Enter value for department: civil
Enter value for marks: 75
old   1: insert into student21 values('&student_id','&student_name','&department','&marks')
new   1: insert into student21 values('3','radha','civil','75')

1 row created.

SQL> /
Enter value for student_id: 4
Enter value for student_name: tara
Enter value for department: IT
Enter value for marks: 85
old   1: insert into student21 values('&student_id','&student_name','&department','&marks')
new   1: insert into student21 values('4','tara','IT','85')

1 row created.

SQL> /
Enter value for student_id: 5
Enter value for student_name: babu
Enter value for department: mech
Enter value for marks: 92
old   1: insert into student21 values('&student_id','&student_name','&department','&marks')
new   1: insert into student21 values('5','babu','mech','92')

1 row created.

SQL> select * from student21;

STUDENT_ID STUDENT_NAME                    DEPARTMENT                MARKS
---------- ------------------------------ -------------------- ----------
         1 maya                           cse                          90
         2 ali                            ece                          95
```

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         3 radha                          civil                          75
         4 tara                           IT                             85
         5 babu                           mech                           92

SQL> select count(*) as total_students from student;
select count(*) as total_students from student
                                       *
ERROR at line 1:
ORA-00942: table or view does not exist


SQL> select count(*) as total_students from student21;

TOTAL_STUDENTS
--------------
             5

SQL> select sum(marks) as total_marks from student21;

TOTAL_MARKS
-----------
        437

SQL> select avg(marks) as average_marks from student21;

AVERAGE_MARKS
-------------
         87.4

SQL> select avg(marks) as total_marks from student21;

TOTAL_MARKS
-----------
       87.4

SQL> select min(marks) as total_marks from student21;

TOTAL_MARKS
-----------
         75

SQL> select min(marks) as min_marks from  student21;

 MIN_MARKS
----------
        75

SQL> select max(marks) as total_marks from student21;p
  2  select max(marks) as total_marks from student21;
select max(marks) as total_marks from student21;p
                                                 *
ERROR at line 1:
ORA-00911: invalid character


SQL> select max(marks) as total_marks from student21;

TOTAL_MARKS
-----------
         95

SQL> select max(marks) as highest
  2
SQL>
SQL> select max(marks) as highest_marks from student21;

HIGHEST_MARKS
-------------
           95

SQL> select department,avg(marks) as avg_marks from student21 group by department;

DEPARTMENT            AVG_MARKS
-------------------- ----------
IT                           85
cse                          90
civil                        75
```

```
ece                          95
mech                         92

SQL>
SQL> /

DEPARTMENT            AVG_MARKS
-------------------- ----------
IT                           85
cse                          90
civil                        75
ece                          95
mech                         92

SQL> insert into student21 values('&student_id','&student_name','&department','&marks');
Enter value for student_id: 6
Enter value for student_name: lata
Enter value for department: cse
Enter value for marks: 80
old   1: insert into student21 values('&student_id','&student_name','&department','&marks')
new   1: insert into student21 values('6','lata','cse','80')

1 row created.

SQL> /
Enter value for student_id: 7
Enter value for student_name: komal
Enter value for department: ece
Enter value for marks: 97
old   1: insert into student21 values('&student_id','&student_name','&department','&marks')
new   1: insert into student21 values('7','komal','ece','97')

1 row created.

SQL> select * from student21;

STUDENT_ID STUDENT_NAME                   DEPARTMENT                MARKS
---------- ------------------------------ -------------------- ----------
         1 maya                           cse                          90
         2 ali                            ece                          95
         3 radha                          civil                        75
         4 tara                           IT                           85
         5 babu                           mech                         92
         6 lata                           cse                          80
         7 komal                          ece                          97

7 rows selected.

SQL> select department,avg(marks) as avg_marks from student21 group by department;

DEPARTMENT            AVG_MARKS
-------------------- ----------
IT                           85
cse                          85
civil                        75
ece                          96
mech                         92

SQL> select department,avg(marks) as avg_marks from student21;
select department,avg(marks) as avg_marks from student21
       *
ERROR at line 1:
ORA-00937: not a single-group group function


SQL> select department,avg(marks0 as total_words from student21;
select department,avg(marks0 as total_words from student21
                          *
ERROR at line 1:
ORA-00907: missing right parenthesis


SQL> select department,avg(marks) as avg_marks from student21 group by department;

DEPARTMENT            AVG_MARKS
-------------------- ----------
IT                           85
```

```
cse                          85
civil                        75
ece                          96
mech                         92

SQL> select department,avg(marks) as total_marks from student21 group by department;

DEPARTMENT           TOTAL_MARKS
-------------------- -----------
IT                            85
cse                           85
civil                         75
ece                           96
mech                          92

SQL> select department,count(marks) as count_student from student21 group by department;

DEPARTMENT           COUNT_STUDENT
-------------------- -------------
IT                               1
cse                              2
civil                            1
ece                              2
mech                             1

SQL> select department,max(marks) as max_marks from student21 group by department;

DEPARTMENT            MAX_MARKS
-------------------- ----------
IT                           85
cse                          90
civil                        75
ece                          97
mech                         92

SQL> select department,avg(marks) as avg_marks from student21 group by department having avg(marks) > 85;

DEPARTMENT            AVG_MARKS
-------------------- ----------
ece                          96
mech                         92

SQL> select department,count(*) as total_students from student21 group by department having count(*)>1;

DEPARTMENT           TOTAL_STUDENTS
-------------------- --------------
cse                               2
ece                               2

SQL> select department,count(*) as no_of_students,avg(marks) as avg_marks,max(marks) as max_marks) from st
    udent21 group by department having avg(marks)>80;
select department,count(*) as no_of_students,avg(marks) as avg_marks,max(marks) as max_marks) from student
    21 group by department having avg(marks)>80
                                                                                                *
ERROR at line 1:
ORA-00923: FROM keyword not found where expected


SQL> select department,count(*) as no_of_students,avg(marks) as avg_marks,max(marks) as max_marks from stu
    dent21 group by department having avg(marks)>80;

DEPARTMENT           NO_OF_STUDENTS  AVG_MARKS  MAX_MARKS
-------------------- -------------- ---------- ----------
IT                                1         85         85
cse                               2         85         90
ece                               2         96         97
mech                              1         92         92

SQL> select * from student21;

STUDENT_ID STUDENT_NAME                   DEPARTMENT                MARKS
---------- ------------------------------ -------------------- ----------
         1 maya                           cse                          90
         2 ali                            ece                          95
         3 radha                          civil                        75
         4 tara                           IT                           85
         5 babu                           mech                         92
```

```
         6 lata                           cse                            80
         7 komal                          ece                            97

7 rows selected.

SQL> create table employee10{
  2  emp_id number(5) primary key,
  3  emp_name varchar2(30),
  4  salary number(8)
  5  );
create table employee10{
                       *
ERROR at line 1:
ORA-00922: missing or invalid option


SQL> create table employee10(
  2  emp_id number(5) primary key,
  3  emp_name varchar2(30),
  4  salary number(8)
  5  );

Table created.

SQL> insert into employee10 values(101,'gopi',100000);

1 row created.

SQL> insert into employee10 values(102,'sanu',50000);

1 row created.

SQL> grant select on employee10 to scott;

Grant succeeded.

SQL> grant select,insert,update on employee to scott;

Grant succeeded.

SQL> revoke insert on employee from scott;

Revoke succeeded.

SQL> insert into employee10 values(103,'sachin',200000);

1 row created.

SQL> commit;

Commit complete.

SQL> insert into employee10 values(104,'murali',105000);

1 row created.

SQL> savepoint sp1;

Savepoint created.

SQL> insert into employee10 values(105,'prakash',100000);

1 row created.

SQL> rollback to sp1;

Rollback complete.

SQL> select * from employee10;

    EMP_ID EMP_NAME                         SALARY
---------- ------------------------------ ----------
       101 gopi                           100000
       102 sanu                            50000
       103 sachin                         200000
       104 murali                         105000
SQL> grant select on employee10 to user_name;
```

```
grant select on employee10 to user_name
                              *
ERROR at line 1:
ORA-01917: user or role 'USER_NAME' does not exist


SQL> create table alumni(
  2  student_name varchar(30)
  3  );

Table created.

SQL> insert into alumni values('maya');

1 row created.

SQL> insert into alumni values('ali');

1 row created.

SQL> insert into alumni values('radha');

1 row created.

SQL> select * from student21 where department in ('cse','IT');

STUDENT_ID STUDENT_NAME                   DEPARTMENT               MARKS
---------- ------------------------------ -------------------- ----------
         1 maya                           cse                          90
         4 tara                           IT                           85
         6 lata                           cse                          80

SQL> select * from student21 where student_id in(1,2);

STUDENT_ID STUDENT_NAME                   DEPARTMENT               MARKS
---------- ------------------------------ -------------------- ----------
         1 maya                           cse                          90
         2 ali                            ece                          95

SQL> select * from student21 where marks > any (
  2  select marks from student where department = 'ece' );
select marks from student where department = 'ece' )
                 *
ERROR at line 2:
ORA-00942: table or view does not exist


SQL> select * from student21 where marks > any (
  2  select marks from student21 where department = 'ece' );

STUDENT_ID STUDENT_NAME                   DEPARTMENT               MARKS
---------- ------------------------------ -------------------- ----------
         7 komal                          ece                          97

SQL> select * from student21 where marks > any (select marks from student21 where department='cse');

STUDENT_ID STUDENT_NAME                   DEPARTMENT               MARKS
---------- ------------------------------ -------------------- ----------
         7 komal                          ece                          97
         2 ali                            ece                          95
         5 babu                           mech                         92
         1 maya                           cse                          90
         4 tara                           IT                           85
SQL>
```